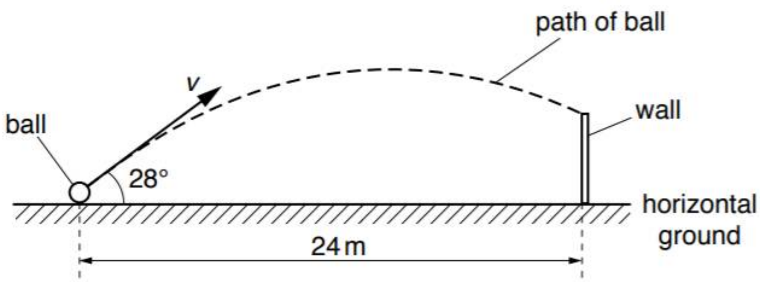
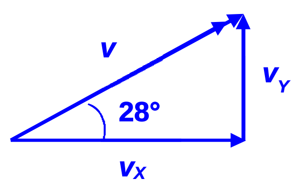
Answer **all** the questions in the spaces provided.

1	**(a)**	A ball is kicked from horizontal ground towards a vertical wall, as shown in Fig. 1.1.  **Fig. 1.1** (not to scale) The horizontal distance between the initial position of the ball and the base of the wall is 24 m. The ball is kicked with an initial velocity v at an angle of 28° above the horizontal. The ball hits the wall after a time of 1.50 s. Air resistance is negligible.		
		(i)	Calculate the initial horizontal component v_X of the velocity of the ball.	
			v_X = ... m s^{-1}	[1]
			$v_X = \frac{s_X}{t} = \frac{24}{1.50} = 16 \text{ m s}^{-1}$	**1**
		(ii)	Show that the initial vertical component v_Y of the velocity of the ball is 8.5 m s^{-1}.	
				[1]
			 **tan 28° = v_Y / v_X** **$v_Y = v_X$ tan 28° = (16) tan 28° = 8.5 m s^{-1} (shown)** **OR** **v_X = v cos 28° --- (1)** **v_Y = v sin 28° --- (2)** **(2)/(1): tan 28° = v_Y / v_X** **$v_Y = v_X$ tan 28° = (16) tan 28° = 8.5 m s^{-1} (shown)**	**1** **(1)**
		(iii)	Calculate the time taken for the ball to reach its maximum height above the ground.	
			time = ... s	[2]

Use $v_{Y,\,final} = v_{Y,\,initial} + a_Y t$ --- (1)

Take vectors upwards as positive.

At maximum height, $v_{Y,\,final} = 0$

Free fall, so $a_Y = -9.81$ m s^{-2}

$v_{Y,\,initial} = v_Y$ from (a)(ii) = +8.5 m s^{-1}

Sub into (1): $0 = 8.5 + (-9.81)t$ **1**

$t = 8.5 / 9.81 = 0.86646 = 0.87$ s (2 s.f.) **1**

(iv) The ball is kicked at time $t = 0$. Assume that the vertical component v_Y of the velocity of the ball is positive in the upwards direction. On Fig. 1.2, sketch the variation with time t of v_Y for the time until the ball hits the wall. **Label this graph Q**.

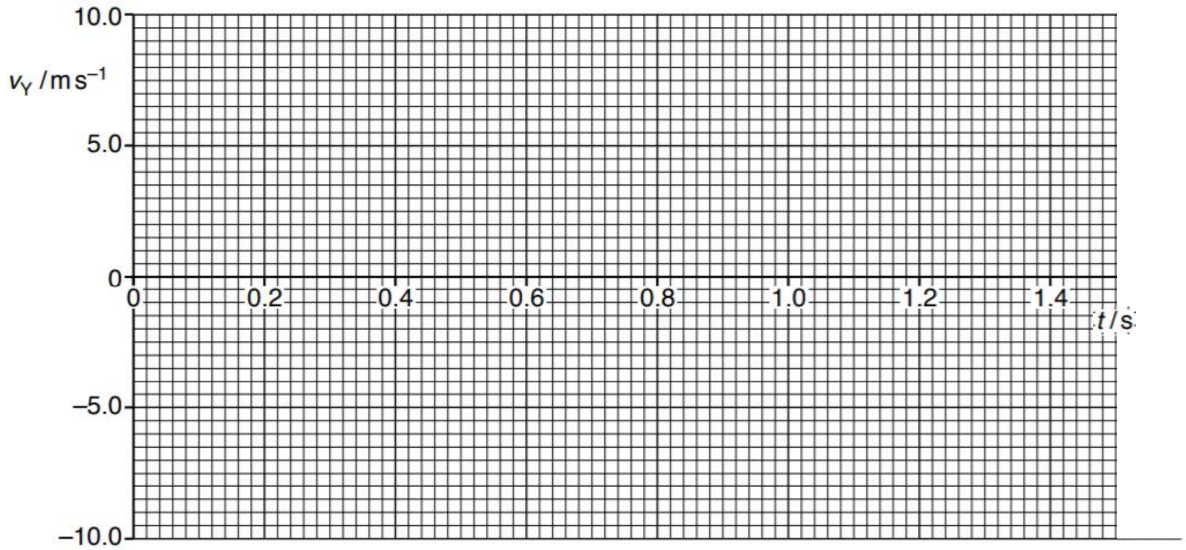
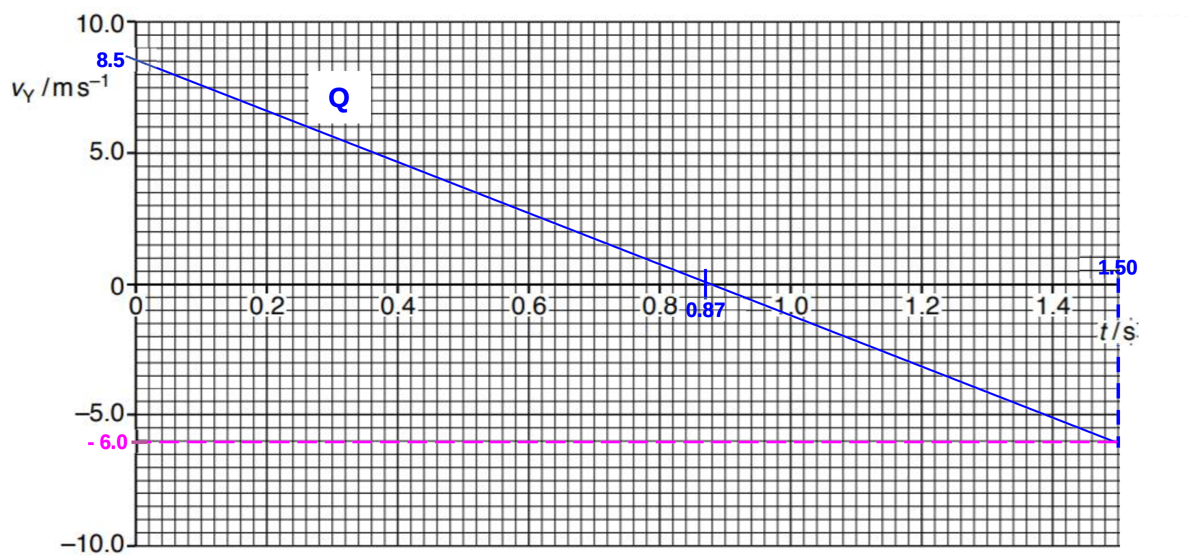


Fig. 1.2 [1]

Solution:



- **Straight line** from **(0, +8.5)** and crosses t-axis at **(0.87, 0).** **1**

(v) Use your graph in Fig. 1.2 to estimate the height of the wall.

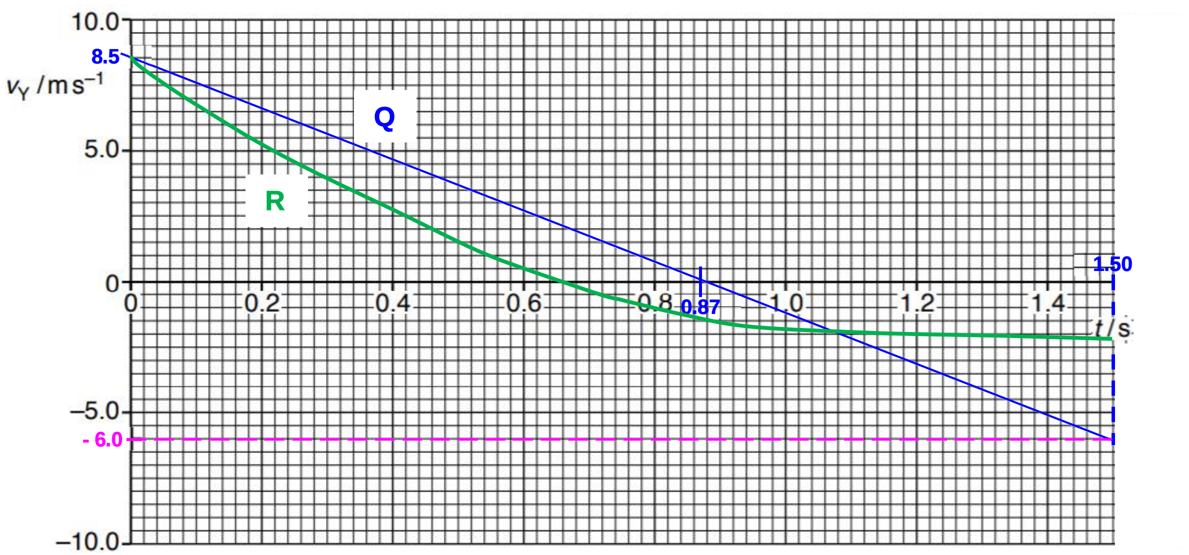
height of wall = ... m [2]

Height of wall = Net change in displacement of the ball
= Total area bounded by the graph and t-axis
= ½ (0.87)(8.5) – ½ (6.0)(1.50-0.87) **M1**
= 1.9075 = 1.9 m (2 s.f.)

A1

(b) On Fig. 1.2, sketch a possible v_Y against t graph if air resistance acts on the ball. Assume the ball does not reach the ground before 1.50s. **Label this graph R.** [2]

Solution:



- Reaches max height earlier, so $v_Y = 0$ at earlier t ($t < 0.87$ s)
- Before max height, greater magnitude of acceleration, so steeper gradient (more negative)
- At v_Y=0, gradients of both graphs are equal.
- After max height, smaller magnitude of acceleration, so gentler gradient (less negative but not zero).
- Net area under graph should be estimated less than net area under original graph, since ball hits wall at a lower height.
- Area under graph before max. height reached is greater than or equal to area under graph after max height reached (since ball either hit the wall at its base, or some distance above its base).

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1